What Holds the Machine

Index Locking, the Decorated Hole, and the Symmetric Channel of the Resolved–Null Split

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Abstract

What pins the resolved–null split in place, what happens if the hole is forced to absorb the resolved sector, and can the hole read the resolved sector in return? This paper answers all three in established mathematics. First, the split is held by an *index*: for $J : \Theta \to O$ the Fredholm index ind $J = \dim \Theta - \dim O$ is fixed, so the observation-space hole (the cokernel, our null sector) and the parameter-space hole (the kernel, the unidentifiable directions) are locked together—forcing the hole to absorb one resolved direction forces one parameter to become unidentifiable. You cannot grow the hole for free. Second, we place this hole against the standard theory of holes: our null sector *is* the cokernel of J, and at first order it behaves exactly as the standard theory says—dual to the kernel, index-locked, invisible to the pseudo-inverse—but it is *decorated* with a covariance and a coupling, and every novel behaviour we found (the Schur floor, the fibre split, the recovery reach, the stratification, the reading) lives in that second-order decoration. Third, the reading runs both ways: the hole reads the resolved sector through $\mathbb{E}[y_R \mid y_N] = C\Sigma_N^{-1} y_N$, with exactly the same canonical correlations as the forward direction, so the geometry between the sectors is symmetric—the asymmetry we exploit is purely *observational*, that we see the resolved sector and not the hole. The reverse reading is not idle: it is a consistency test on any external null estimate, predicting the resolved data a null model implies and comparing it to what is observed. Finally, the split is rigid: $R = \mathrm{Im}(J)$ is invariant under reparametrization of the parameter space and moves only when the observation metric changes. All statements are verified. What holds the machine is thus the index (locking the two holes), the coupling (a symmetric channel between the sectors), and the metric (fixing the split); forcing the hole to absorb the resolved sector drags an equal loss of identifiability with it.

Contents

1 Three questions about stability

The corpus has read the hole by understanding the resolved sector. Three questions turn that around and ask what holds the arrangement together. If the hole is forced to grow—to absorb resolved directions—what does it drag with it? How does this hole compare to the standard mathematical theory of a hole (an absence, a cokernel)? And can the hole read its resolved counterpart, reversing the direction of inference we have used throughout? The answers are an index law, a decoration principle, and a symmetry.

2 The decorated hole

Definition 2.1 (The hole is the cokernel)**.** For the observation map $J : \Theta \to O$, the resolved sector is $R = \mathrm{Im}(J)$ and the null sector is $N = R^{\perp_g} \cong \operatorname{coker} J = O/\operatorname{Im}(J)$: the hole is the cokernel of J.

Proposition 2.2 (Standard behaviour at first order)**.** *As a cokernel, N obeys the standard theory of holes: it is dual to the kernel $\ker J$, it is index-locked to it (§3), and it is invisible to the Moore–Penrose pseudo-inverse (J^+ vanishes on N). These are first-order (subspace) facts, identical to any cokernel.*

Remark 2.3 (The decoration is the novelty)*.* Our hole carries more than a bare cokernel: a covariance Σ_N and a coupling C. Every non-standard behaviour established for it—the Schur floor $\Sigma_N \succeq C^\top \Sigma_R^{-1} C$, the coupled/free fibre split, the recovery reach measured by canonical correlations, the Weyl-chamber stratification with its diabolical phases, and the reading of §5—is a property of this second-order decoration, not of the bare cokernel. The standard theory governs the first order; the decoration governs everything new.

3 Index locking: what holds the machine

Theorem 3.1 (The two holes are index-locked)**.** *The index* $\operatorname{ind} J = \dim\ker J - \dim\operatorname{coker} J = \dim\Theta - \dim O$ *is fixed, independent of J. Hence* $\dim\ker J$ *and* $\dim N = \dim\operatorname{coker} J$ *change only together: forcing the observation-space hole to grow by one (absorbing a resolved direction) forces the parameter-space kernel to grow by one (a parameter becomes unidentifiable). The observation-hole cannot grow for free.*

Proof. Rank–nullity: $\dim\ker J = \dim\Theta - \operatorname{rank} J$ and $\dim\operatorname{coker} J = \dim O - \operatorname{rank} J$, so their difference is $\dim\Theta - \dim O$, independent of $\operatorname{rank} J$ [1]. Verified for $J : \mathbb{R}^5 \to \mathbb{R}^7$: as $\operatorname{rank} J$ falls $5 \to 4 \to 3$, $\dim\ker J$ rises $0 \to 1 \to 2$ and $\dim N$ rises $2 \to 3 \to 4$, with $\operatorname{ind} J = -2$ throughout. □

Remark 3.2 (The physical reading of the lock)*.* Absorbing a resolved direction into the hole is exactly a loss of rank in J, and by Theorem 3.1 it costs a direction of parameter identifiability. The resolved/null split in observation space is therefore tied to the identifiable/unidentifiable split in parameter space; one cannot be moved without the other. This index is a first thing holding the machine in place.

4 Reparametrization rigidity

Theorem 4.1 (The resolved sector refuses parameter reparametrization)**.** $R = \mathrm{Im}(J)$ *is invariant under any invertible reparametrization of the parameter space, $J \mapsto JA$:* $\mathrm{Im}(JA) = \mathrm{Im}(J)$*. The null sector $N = R^{\perp_g}$ therefore also refuses parameter reparametrization, and moves only when the observation metric g changes.*

Proof. $\mathrm{Im}(JA) = \mathrm{Im}(J)$ for invertible A (right multiplication permutes the domain without changing the column space). Verified: the projector onto $\mathrm{Im}(J)$ is unchanged under a random $J \mapsto JA$ to 2.8×10^{-15}. The orthogonal complement depends on g but not on A. □

Corollary 4.2 (The resolved/null boundary is a wall). *Absorption happens only at the locus where a singular value of J vanishes—a codimension-one wall in the space of observation maps, dual to the coupled/free wall inside the null sector. Its sharpness is the corresponding spectral gap (Davis–Kahan). The split is thus rigid away from the wall and changes type only on crossing it.*

5 Reverse reading: the symmetric channel

We have read the hole from the resolved sector via $\mathbb{E}[y_N \mid y_R] = C^\top \Sigma_R^{-1} y_R$. The reverse holds.

Theorem 5.1 (The hole reads the resolved sector, symmetrically). *The resolved sector is recovered from the hole by $\mathbb{E}[y_R \mid y_N] = C\Sigma_N^{-1} y_N$, and the reach of this reverse reading is governed by the* same *canonical correlations as the forward reading: the whitened coupling and its transpose share their singular values. The geometry between the sectors is symmetric; the asymmetry the framework exploits is purely observational—that the resolved sector is observed and the hole is not.*

Proof. Both conditional means are the Gaussian linear-MMSE estimators [2]; the canonical correlations of (R, N) equal those of (N, R) (singular values of $K = \Sigma_R^{-1/2} C \Sigma_N^{-1/2}$ and of $K^\top$ coincide). Verified: forward and reverse canonical correlations agree to 2.8×10^{-17}. □

Corollary 5.2 (Reverse reading is a consistency test). *Because the hole is not observed, the reverse reading is used not to recover R (which is observed) but to test an external null estimate: a proposed (Σ_N, C) predicts the resolved data $\mathbb{E}[y_R \mid y_N] = C\Sigma_N^{-1} y_N$ it implies, and comparison with the observed resolved data is a consistency check on the estimate. The reverse channel is therefore the validation counterpart of the acceptance theory: what the hole "reads" about the resolved sector is exactly what a correct null model must reproduce.*

6 What holds the machine

Three things hold the split in place, and they are distinct.

1. **The index** (Theorem 3.1) locks the observation-hole to the parameter-kernel: the hole cannot absorb the resolved sector without an equal loss of identifiability.
2. **The coupling** (Theorem 5.1) ties the sectors by a symmetric bidirectional channel whose capacity in both directions is the canonical correlations—the same invariant that rigidifies the automorphisms and classifies the strata.
3. **The metric** (Theorem 4.1) fixes the split: $R = \mathrm{Im}(J)$ is rigid under parameter reparametrization, and only a change of observation metric moves the boundary, which otherwise changes type only at a codimension-one wall.

Forcing the hole to absorb the resolved sector is therefore not a free deformation: it is a rank loss, it drags an equal loss of parameter identifiability, it occurs only at a wall, and it leaves the reverse channel intact to certify what a null model must reproduce.

7 Scope and boundaries

Non-claim 7.1. All statements are established mathematics: the Fredholm index and rank–nullity; the four fundamental subspaces and the Moore–Penrose pseudo-inverse; Gaussian linear-MMSE estimation and the symmetry of canonical correlations; and the invariance of the column space under domain reparametrization. The novelty is the synthesis—identifying the index, the coupling, and the metric as the three things that hold the resolved/null split, framing the null sector as a decorated cokernel, and using the reverse reading as a consistency test. The reverse reading is geometrically exact but epistemically bounded: because the hole is unobserved, its use is validation, not recovery of already-observed data. No physical claim is made.

8 Conclusion

The hole of a resolution geometry is the cokernel of the observation map, and at first order it behaves as the standard theory of holes dictates: dual to the kernel, index-locked, blind to the pseudo-inverse. What is new lives in its second-order decoration—covariance and coupling—and it is that decoration that gives the floor, the split, the reach, the stratification, and the reading. The arrangement is held by three things: an index that forbids the hole from absorbing the resolved sector without an equal loss of parameter identifiability, a coupling that connects the sectors by a channel symmetric in both directions, and a metric that makes the resolved sector rigid under reparametrization. And the inference runs both ways: the hole can read its resolved counterpart with the same fidelity the resolved sector reads it, so that the direction we have always taken is a choice made by observation, not by geometry—and the reverse direction, though we cannot harvest it for recovery, is exactly the test any honest null estimate must pass.

References

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